Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology

B.Tech. | Sem-4 | Winter-2023

Course Code	:	2101CS403	**Date**	:	20-10-2023
Course Name	:	Operating System	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	What is system call? Explain steps for system call execution.	**4**
	(B)	Give the view of OS as a resource manager.	**3**
		OR	
		Write the short note on distributed OS.	
	(C)	Explain the types of Operating System.	**7**
		OR	
		List out the types of architecture/ structure of OS. Explain Layered and Monolithic Operating Systems.	
Q.2	**(A)**	Explain the PCB with block diagram.	**4**
	(B)	Discuss context switching in OS.	**3**
		OR	
		Differentiate the thread and process.	
	(C)	Explain the shortest remaining time next scheduling algorithm in short. For the process P0,P1,P2,P3 arrival time is 0,1,3,5 and burst time is 10,6,2,4 respectively. Apply SRTN algorithm on it and draw Gaant chart, calculate average turn around time and average waiting time.	**7**
		OR	
		Explain the shortest job first scheduling algorithm in short. For the process P0,P1,P2,P3 arrival time is 0,1,3,5 and burst time is 10,6,2,4 respectively. Apply SJF algorithm on it and draw Gaant chart, calculate average turn around time and average waiting time.	
Q.3	**(A)**	What is deadlock? which four conditions must be present for a deadlock to occur? explain it?	**4**
	(B)	Explain Test and set lock mechanism in IPC.	**3**
		OR	
		Explain race condition and mutual exclusion.	
	(C)	Describe Producer - Consumer problem using semaphore in IPC.	**7**

OR

Describe Bankers algorithm with suitable example.

Q.4	**(A)**	Discuss paging with example.	**4**
	(B)	Explain Virtual Memory with the help of diagram.	**3**

OR

Describe Fixed and variable memory allocation in short.

	(C)	Calculate the page fault rates for below reference string in case of FIFO and Optimal page replacement algorithm. Assume the memory size is 3 page frames and all frames are initially empty. 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 0, 2, 0, 1, 7, 0, 1.	**7**

OR

Calculate the page fault rates for below reference string in case of LRU and Optimal page replacement algorithm. Assume the memory size is 3 page frames and all frames are initially empty. 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 0, 2, 0, 1, 7, 0, 1.

Q.5	**(A)**	Explain different File operations.	**4**
	(B)	Write short note on Direct Memory Access.	**3**

OR

Explain different types of Files.

	(C)	Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for the C-Look disk scheduling Algorithms?	**7**

OR

Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9, and 12, in that order.
Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for the C-SCAN disk scheduling Algorithms?

***